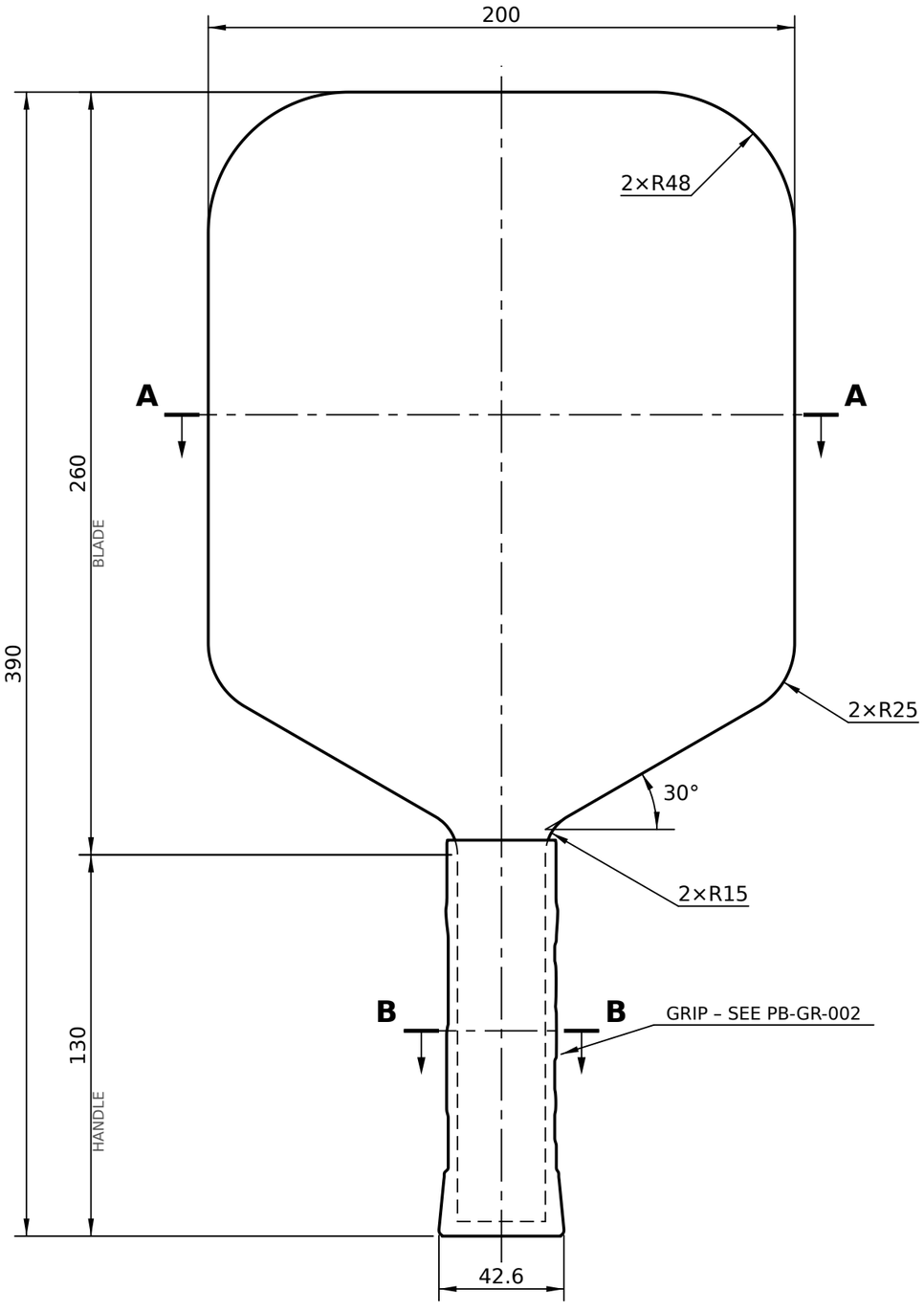
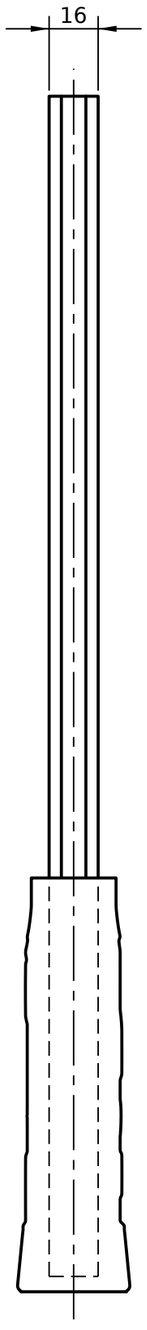
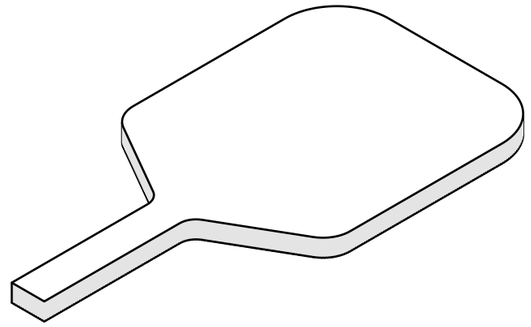


FRONT VIEW



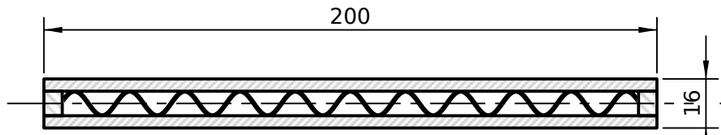
SIDE VIEW

LAY-UP 4 / 8 / 4



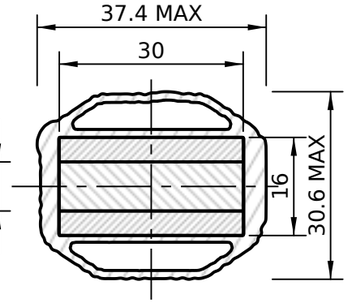
ISOMETRIC VIEW

SCALE 1:4 (BODY ONLY)



SECTION A–A

SCALE 1:2 · CF 4 / WAVE-RIB CORE 8 / CF 4 · CORE DETAIL: SHEET 3



SECTION B–B

SCALE 1:1 · GRIP: PB-GR-002

NOTES

1. ALL DIMENSIONS IN MILLIMETRES. DO NOT SCALE THE DRAWING. UNLESS STATED: LINEAR ±0.5, ANGULAR ±0.5°.
2. PROFILE SYMMETRICAL ABOUT THE CENTRE LINE. ALL ARCS TANGENT TO ADJOINING EDGES.
3. SANDWICH: CF FACE SHEET 4 + WAVE-RIB CORE 8 + CF FACE SHEET 4 = 16 ±0.3, UNIFORM OVER BLADE AND HANDLE. BODY (SHEETS + CORE) FROM y = 5 TO THE TIP; GRIP BASE y = 0 … 5.
4. FACE SHEETS (2 OFF, IDENTICAL): WOVEN CARBON/EPOXY PLATE 4 THK, CUT TO PB-EL-001_profile.dxf, OUTER EDGES R0.8. TEMPLATE: SHEET 2.
5. RIB SUPPORT (CORE): 3D-PRINTED PA-CF, CLOSED FRAME 6, WAVE RIB λ18 / WALL 1.6 / LANDS 4.25, SOLID TO y = 150 WITH 45° FLUTE RUN-OUT – SEE SHEET 3 (1- OR 2-PIECE PRINT).
6. BOND CORE TO SHEETS: STRUCTURAL EPOXY, ABRADED + PRIMED SURFACES, FILLET ALL LAND EDGES (CRITICAL ITEM – SEE THE STRESS REPORT).
7. GRIP: PRINTED PLA GRIP (ANY OF THE TEN REV-8 DESIGNS; DRAWN: CARBON MATRIX, 61 g), SLOT 30.2 × 16 × 130 DEEP, LOCKED BY A DIN 7984 M3 × 12 SCREW INTO AN M3 HEAT-SET INSERT – SEE PB-GR-002.
8. USA PICKLEBALL LIMITS: L + W = 590 ≤ 609.6 (24 in), L = 390 ≤ 431.8 (17 in), INCL. GRIP. AN EDGE GUARD COUNTS TOWARDS THESE LIMITS (MARGIN 19.6).
9. MASS 849 g WITH THE GRIP, BALANCE POINT 236 FROM THE BUTT (FROM THE 3D MODEL).

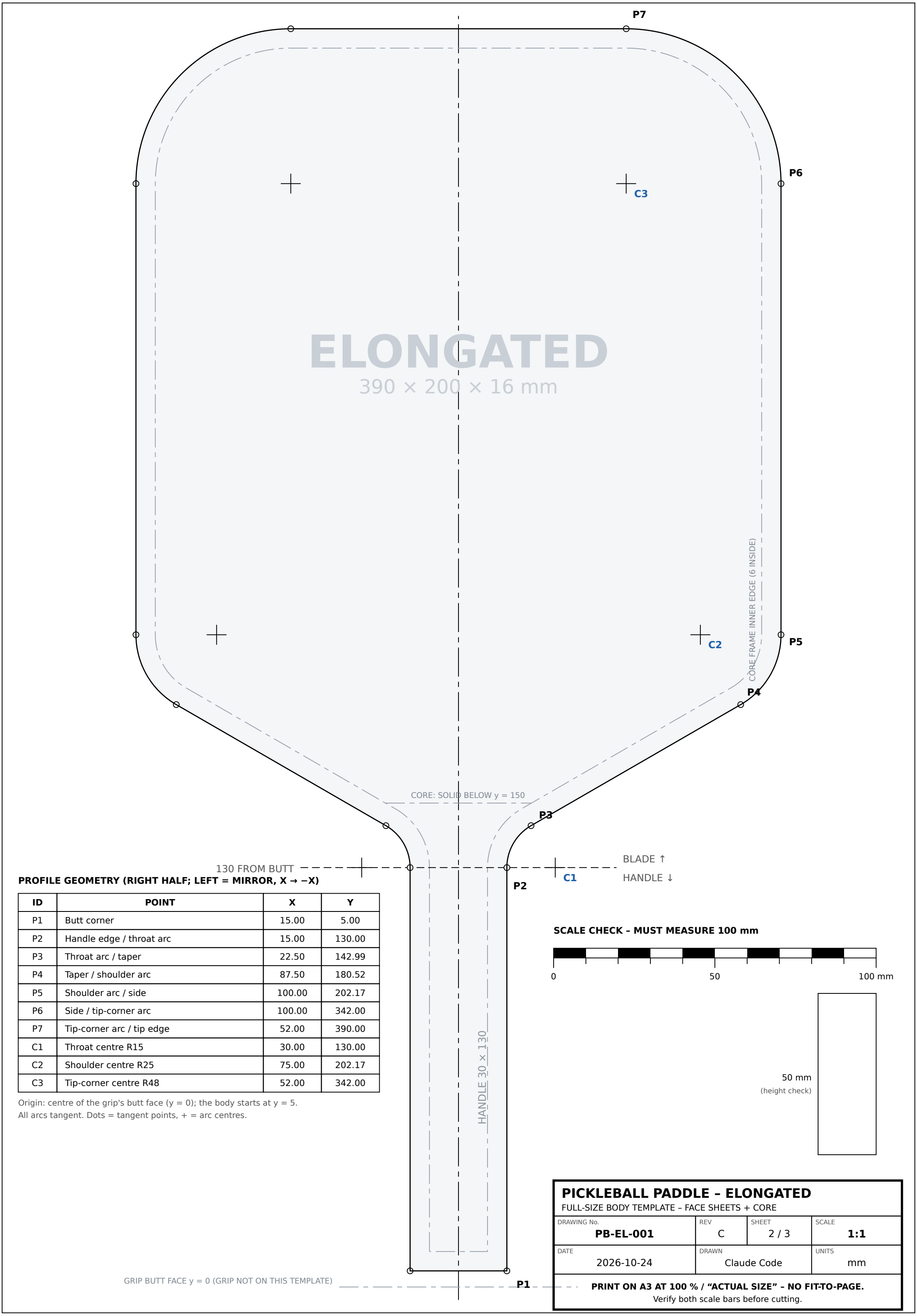
PICKLEBALL PADDLE – ELONGATED

GENERAL ARRANGEMENT · 390 × 200 × 16 mm · CF / WAVE-RIB SANDWICH

DRAWN	DATE	CHECKED	APPROVED
Claude Code	2026-10-24		

DRAWING No.	REV	SHEET	SIZE	SCALE	UNITS
PB-EL-001	C	1 / 3	A3	1:2	mm

PROJECTION	MATERIAL	STATUS
	SEE NOTES	ISSUED FOR REVIEW



PROFILE GEOMETRY (RIGHT HALF; LEFT = MIRROR, X → −X)

ID	POINT	X	Y
P1	Butt corner	15.00	5.00
P2	Handle edge / throat arc	15.00	130.00
P3	Throat arc / taper	22.50	142.99
P4	Taper / shoulder arc	87.50	180.52
P5	Shoulder arc / side	100.00	202.17
P6	Side / tip-corner arc	100.00	342.00
P7	Tip-corner arc / tip edge	52.00	390.00
C1	Throat centre R15	30.00	130.00
C2	Shoulder centre R25	75.00	202.17
C3	Tip-corner centre R48	52.00	342.00

Origin: centre of the grip's butt face (y = 0); the body starts at y = 5.
All arcs tangent. Dots = tangent points, + = arc centres.

SCALE CHECK – MUST MEASURE 100 mm

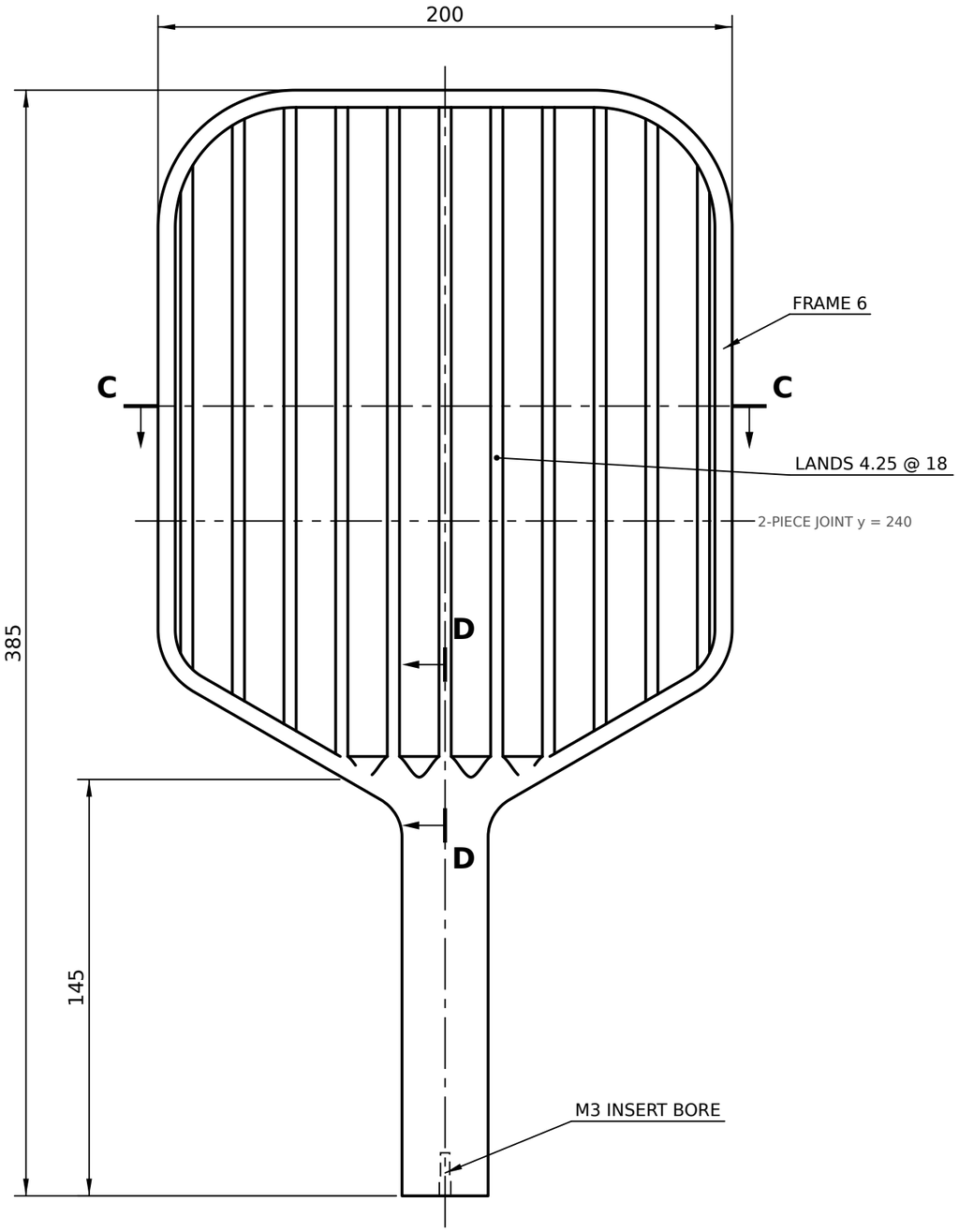
PICKLEBALL PADDLE – ELONGATED

FULL-SIZE BODY TEMPLATE – FACE SHEETS + CORE

DRAWING No.	REV	SHEET	SCALE
PB-EL-001	C	2 / 3	**1:1**

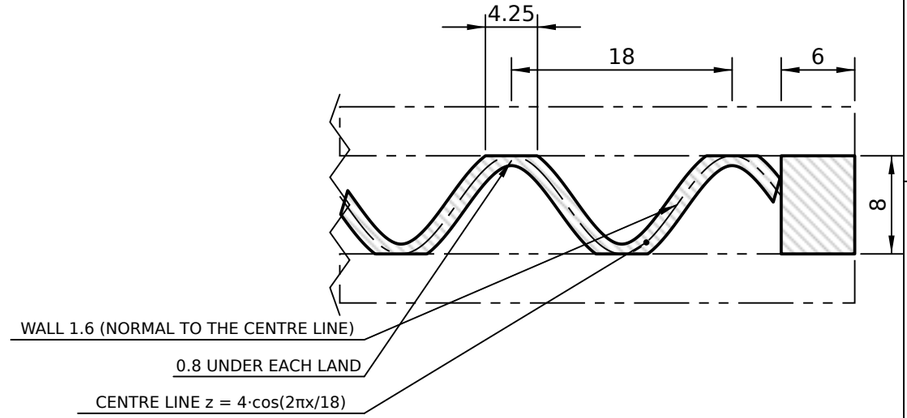
DATE	DRAWN	UNITS
2026-10-24	Claude Code	mm

PRINT ON A3 AT 100 % / “ACTUAL SIZE” – NO FIT-TO-PAGE.
Verify both scale bars before cutting.



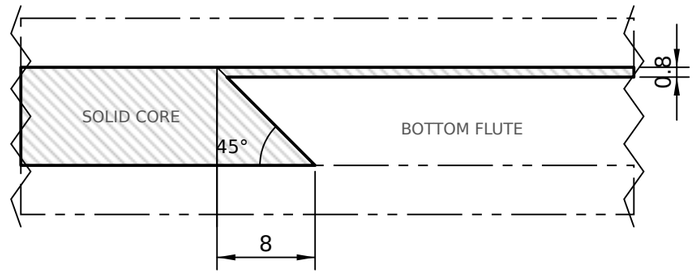
PLAN VIEW (TOP FACE)

SCALE 1:2 · BOTTOM LANDS OFFSET ½ PITCH



SECTION C–C (PART)

SCALE 2:1 · FACE SHEETS SHOWN PHANTOM



SECTION D–D (PART)

SCALE 2:1 · ON THE CENTRE LAND (x = 0), RUN-OUT FROM y = 150; TOP FLUTES MIRROR IT

NOTES

1. MATERIAL: PA-CF (CARBON-FIBRE-FILLED NYLON), E ≈ 4 GPa, ρ 1.18 g/cm³. DRY THE FILAMENT BEFORE PRINTING.
2. PRINT FLAT, 100 % SOLID: WALL 1.6 = 4 PERIMETERS (0.4 NOZZLE), 0.2 LAYERS. NO SUPPORTS (45° RUN-OUTS).
3. THICKNESS 8 ±0.1. BOTH FACES ARE BOND SURFACES, FLAT WITHIN 0.1.
4. WAVE RIB λ18, WALL 1.6, CENTRE-LINE AMPLITUDE 4; CRESTS CUT FLAT = LANDS 4.25 ON BOTH FACES.
5. FRAME 6, CLOSED. SOLID TO y = 150; EACH FLUTE STARTS WITH A 45° RUN-OUT, 8 LONG (D–D).
6. M3 HEAT-SET INSERT BORE Ø4 × 7 + Ø3.2 × 15 IN THE BUTT FACE (GRIP SCREW) – SEE PB-GR-002.
7. BONDING: ABRADE (P120), DEGREASE, STRUCTURAL EPOXY, FILLET EVERY LAND EDGE.
8. PRINT FILES output/rib_support/: elongated_rib_support_1piece.stl (BED ≥ 391 × 206) OR _2piece_A + _B (256 BED): 7 SOLID SPLICE BAND AT y = 240, DOVETAIL JOINT 0.1 CLEARANCE PER SIDE - GLUE IT BEFORE BONDING THE SHEETS.
9. MASS 178 g. y FROM THE GRIP BUTT FACE (y = 0); THE RIB SUPPORT STARTS AT y = 5.

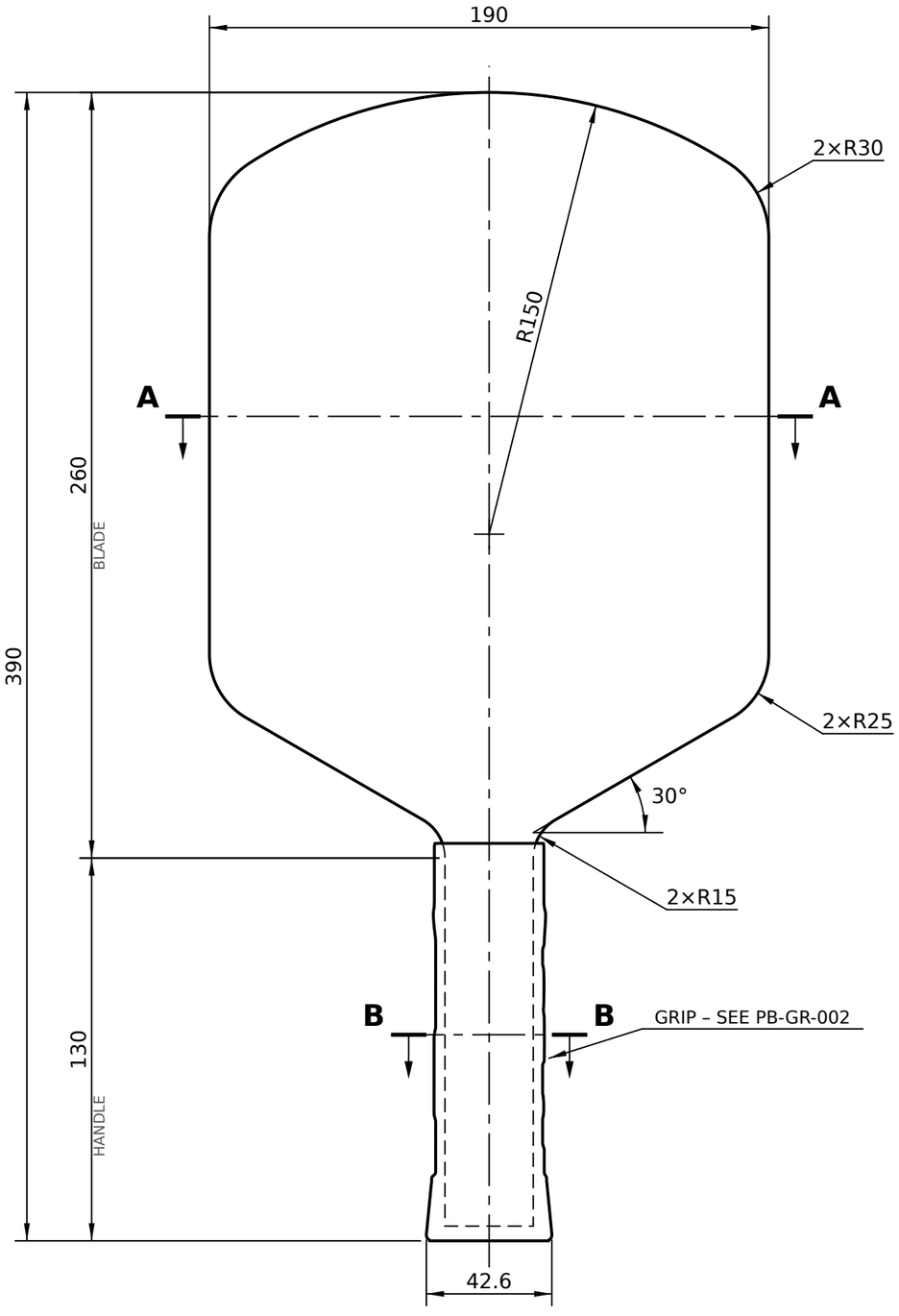
RIB SUPPORT – ELONGATED

RIB SUPPORT (CORE) · 3D-PRINTED PA-CF · 385 × 200 × 8 mm

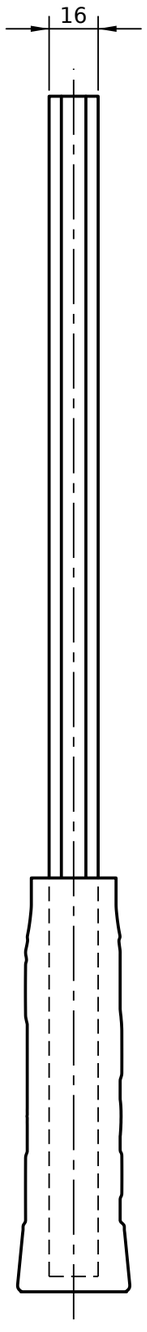
DRAWN	DATE	CHECKED	APPROVED
Claude Code	2026-10-24		

DRAWING No.	REV	SHEET	SIZE	SCALE	UNITS
PB-EL-001	C	3 / 3	A3	1:2, 2:1	mm

PROJECTION	MATERIAL	STATUS
	PA-CF (NOTE 1)	ISSUED FOR REVIEW

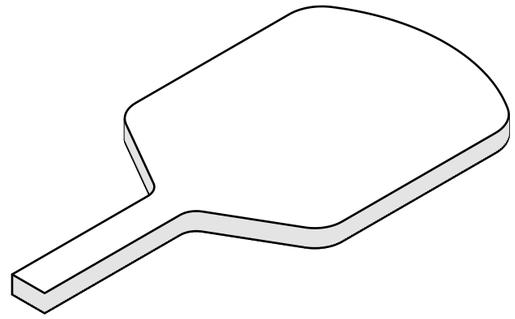


FRONT VIEW



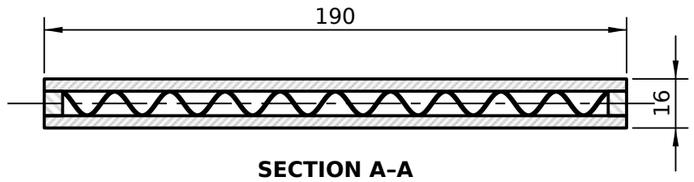
SIDE VIEW

LAY-UP 4 / 8 / 4



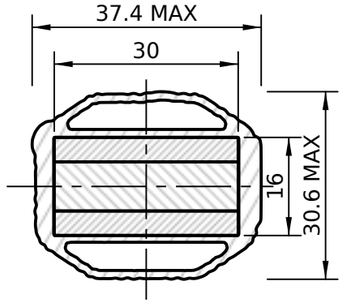
ISOMETRIC VIEW

SCALE 1:4 (BODY ONLY)



SECTION A–A

SCALE 1:2 · CF 4 / WAVE-RIB CORE 8 / CF 4 · CORE DETAIL: SHEET 3



SECTION B–B

SCALE 1:1 · GRIP: PB-GR-002

NOTES

1. ALL DIMENSIONS IN MILLIMETRES. DO NOT SCALE THE DRAWING. UNLESS STATED: LINEAR ±0.5, ANGULAR ±0.5°.
2. PROFILE SYMMETRICAL ABOUT THE CENTRE LINE. ALL ARCS TANGENT TO ADJOINING EDGES.
3. SANDWICH: CF FACE SHEET 4 + WAVE-RIB CORE 8 + CF FACE SHEET 4 = 16 ±0.3, UNIFORM OVER BLADE AND HANDLE. BODY (SHEETS + CORE) FROM y = 5 TO THE TIP; GRIP BASE y = 0 … 5.
4. FACE SHEETS (2 OFF, IDENTICAL): WOVEN CARBON/EPOXY PLATE 4 THK, CUT TO PB-HY-001_profile.dxf, OUTER EDGES R0.8. TEMPLATE: SHEET 2.
5. RIB SUPPORT (CORE): 3D-PRINTED PA-CF, CLOSED FRAME 6, WAVE RIB λ18 / WALL 1.6 / LANDS 4.25, SOLID TO y = 150 WITH 45° FLUTE RUN-OUT – SEE SHEET 3 (1- OR 2-PIECE PRINT).
6. BOND CORE TO SHEETS: STRUCTURAL EPOXY, ABRADED + PRIMED SURFACES, FILLET ALL LAND EDGES (CRITICAL ITEM – SEE THE STRESS REPORT).
7. GRIP: PRINTED PLA GRIP (ANY OF THE TEN REV-8 DESIGNS; DRAWN: CARBON MATRIX, 61 g), SLOT 30.2 × 16 × 130 DEEP, LOCKED BY A DIN 7984 M3 × 12 SCREW INTO AN M3 HEAT-SET INSERT – SEE PB-GR-002.
8. USA PICKLEBALL LIMITS: L + W = 580 ≤ 609.6 (24 in), L = 390 ≤ 431.8 (17 in), INCL. GRIP. AN EDGE GUARD COUNTS TOWARDS THESE LIMITS (MARGIN 29.6).
9. MASS 802 g WITH THE GRIP, BALANCE POINT 230 FROM THE BUTT (FROM THE 3D MODEL).

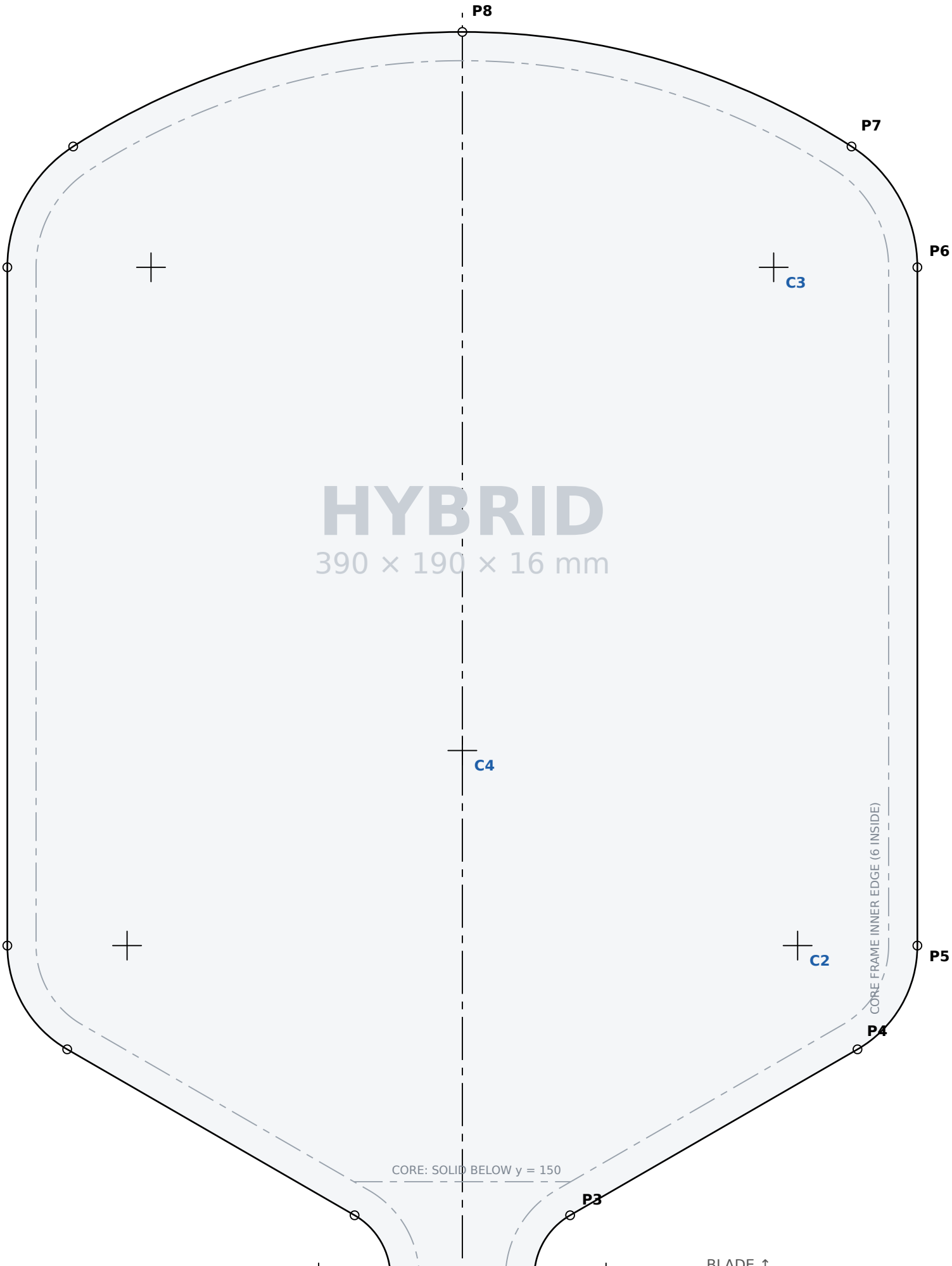
PICKLEBALL PADDLE – HYBRID

GENERAL ARRANGEMENT · 390 × 190 × 16 mm · CF / WAVE-RIB SANDWICH

DRAWN	DATE	CHECKED	APPROVED
Claude Code	2026-10-24		

DRAWING No.	REV	SHEET	SIZE	SCALE	UNITS
PB-HY-001	C	1 / 3	A3	1:2	mm

PROJECTION	MATERIAL	STATUS
	SEE NOTES	ISSUED FOR REVIEW

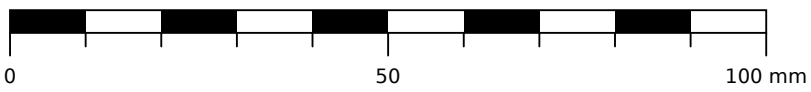


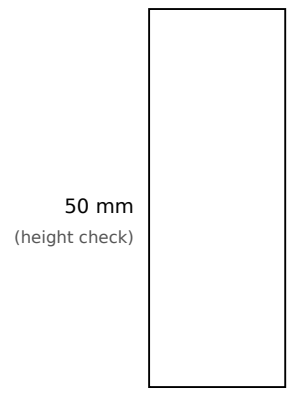
PROFILE GEOMETRY (RIGHT HALF; LEFT = MIRROR, X → −X)

ID	POINT	X	Y
P1	Butt corner	15.00	5.00
P2	Handle edge / throat arc	15.00	130.00
P3	Throat arc / taper	22.50	142.99
P4	Taper / shoulder arc	82.50	177.63
P5	Shoulder arc / side	95.00	199.28
P6	Side / crown-shoulder arc	95.00	340.87
P7	Crown shoulder / dome	81.25	366.09
P8	Apex (on centre line)	0.00	390.00
C1	Throat centre R15	30.00	130.00
C2	Shoulder centre R25	70.00	199.28
C3	Crown-shoulder centre R30	65.00	340.87
C4	Dome centre R150	0.00	240.00

Origin: centre of the grip's butt face (y = 0); the body starts at y = 5.
All arcs tangent. Dots = tangent points, + = arc centres.

SCALE CHECK – MUST MEASURE 100 mm





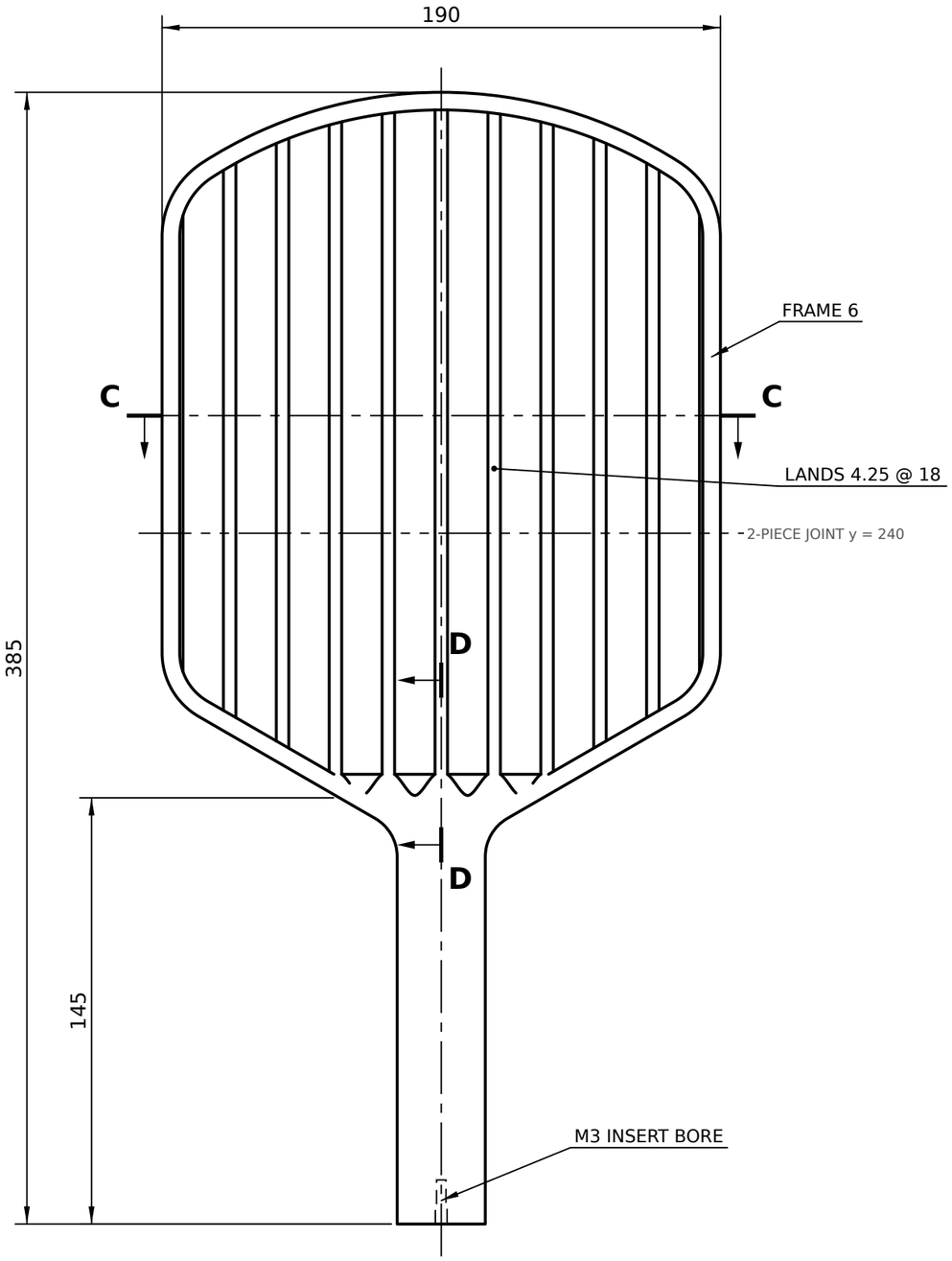
PICKLEBALL PADDLE – HYBRID

FULL-SIZE BODY TEMPLATE – FACE SHEETS + CORE

DRAWING No.	REV	SHEET	SCALE
PB-HY-001	C	2 / 3	**1:1**

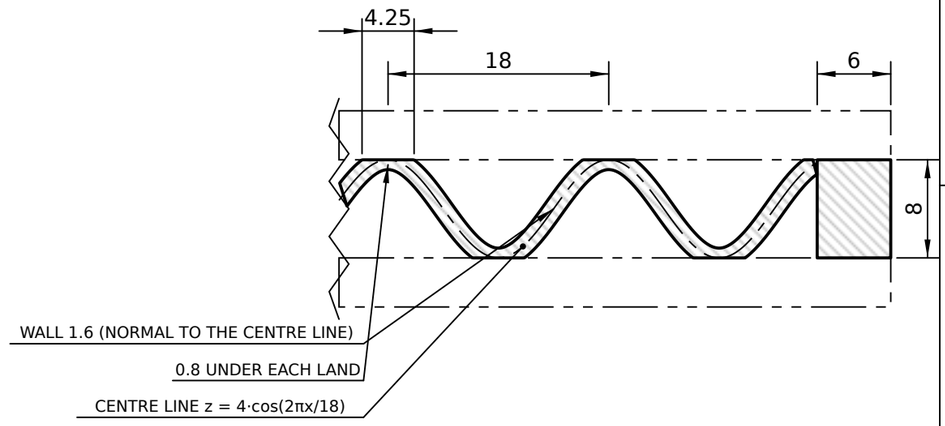
DATE	DRAWN	UNITS
2026-10-24	Claude Code	mm

PRINT ON A3 AT 100 % / “ACTUAL SIZE” – NO FIT-TO-PAGE.
Verify both scale bars before cutting.



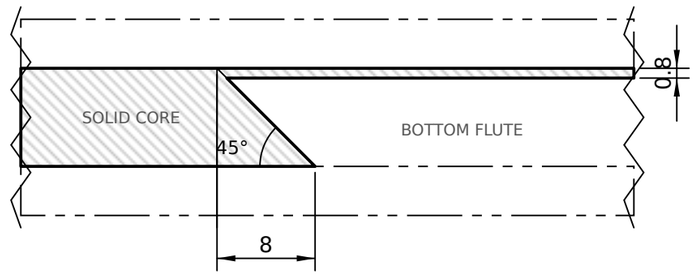
PLAN VIEW (TOP FACE)

SCALE 1:2 · BOTTOM LANDS OFFSET ½ PITCH



SECTION C–C (PART)

SCALE 2:1 · FACE SHEETS SHOWN PHANTOM



SECTION D–D (PART)

SCALE 2:1 · ON THE CENTRE LAND (x = 0), RUN-OUT FROM y = 150; TOP FLUTES MIRROR IT

NOTES

1. MATERIAL: PA-CF (CARBON-FIBRE-FILLED NYLON), E ≈ 4 GPa, ρ 1.18 g/cm³. DRY THE FILAMENT BEFORE PRINTING.
2. PRINT FLAT, 100 % SOLID: WALL 1.6 = 4 PERIMETERS (0.4 NOZZLE), 0.2 LAYERS. NO SUPPORTS (45° RUN-OUTS).
3. THICKNESS 8 ±0.1. BOTH FACES ARE BOND SURFACES, FLAT WITHIN 0.1.
4. WAVE RIB λ18, WALL 1.6, CENTRE-LINE AMPLITUDE 4; CRESTS CUT FLAT = LANDS 4.25 ON BOTH FACES.
5. FRAME 6, CLOSED. SOLID TO y = 150; EACH FLUTE STARTS WITH A 45° RUN-OUT, 8 LONG (D–D).
6. M3 HEAT-SET INSERT BORE Ø4 × 7 + Ø3.2 × 15 IN THE BUTT FACE (GRIP SCREW) – SEE PB-GR-002.
7. BONDING: ABRADE (P120), DEGREASE, STRUCTURAL EPOXY, FILLET EVERY LAND EDGE.
8. PRINT FILES output/rib_support/: hybrid_rib_support_1piece.stl (BED ≥ 391 × 196) OR _2piece_A + _B (256 BED): 7 SOLID SPLICE BAND AT y = 240, DOVETAIL JOINT 0.1 CLEARANCE PER SIDE - GLUE IT BEFORE BONDING THE SHEETS.
9. MASS 170 g. y FROM THE GRIP BUTT FACE (y = 0); THE RIB SUPPORT STARTS AT y = 5.

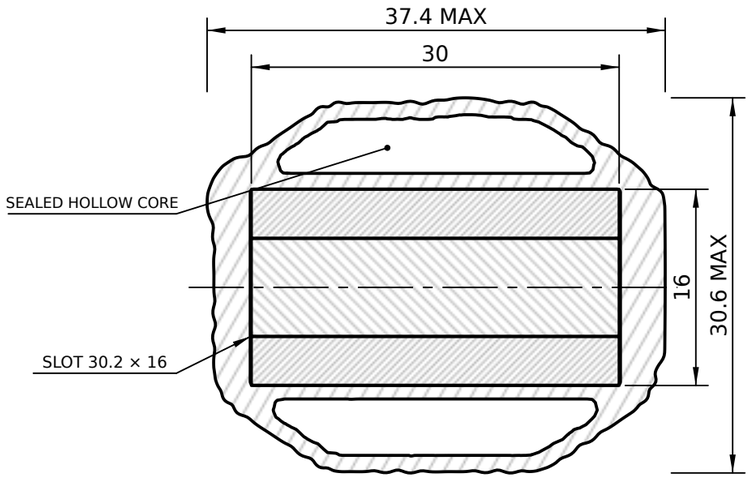
RIB SUPPORT – HYBRID

RIB SUPPORT (CORE) · 3D-PRINTED PA-CF · 385 × 190 × 8 mm

DRAWN	DATE	CHECKED	APPROVED
Claude Code	2026-10-24		

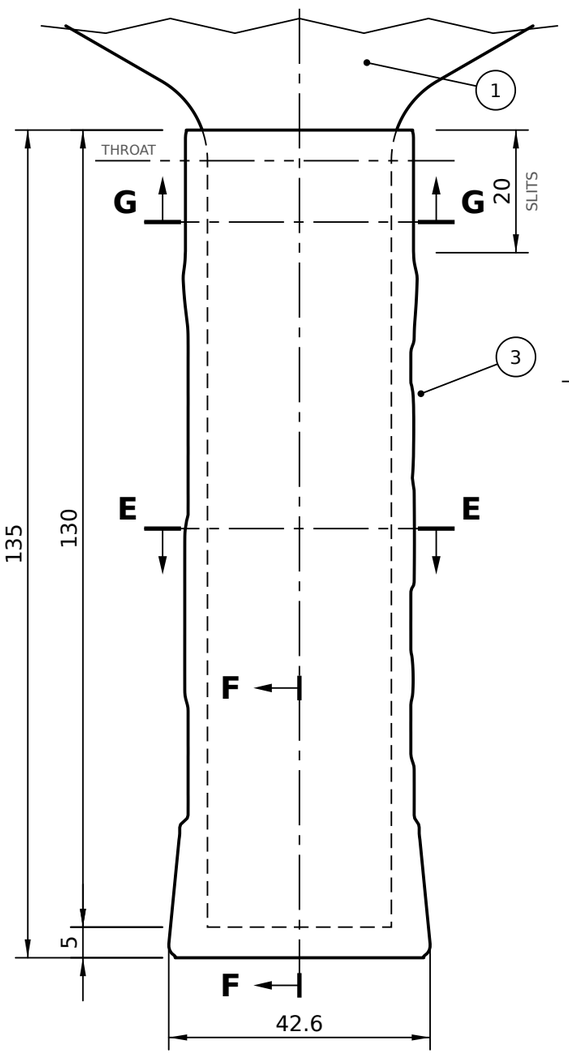
DRAWING No.	REV	SHEET	SIZE	SCALE	UNITS
PB-HY-001	C	3 / 3	A3	1:2, 2:1	mm

PROJECTION	MATERIAL	STATUS
	PA-CF (NOTE 1)	ISSUED FOR REVIEW



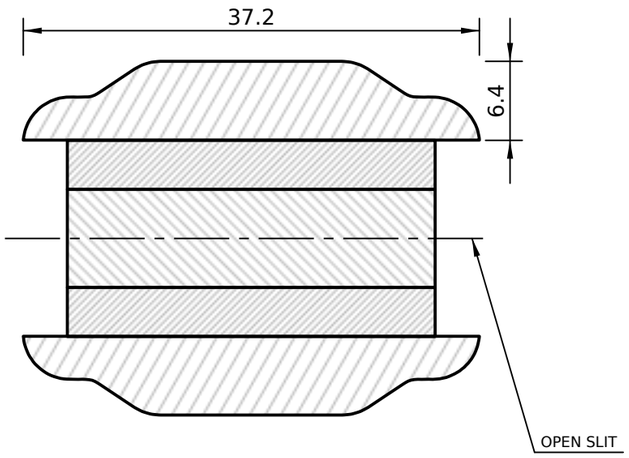
SECTION E–E

SCALE 2:1



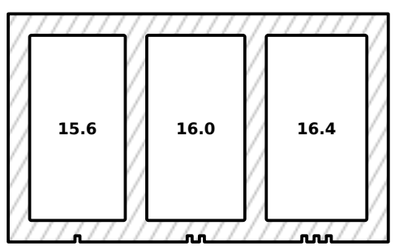
HANDLE IN THE GRIP (FACE VIEW)

SCALE 1:1 · DRAWN: CARBON MATRIX



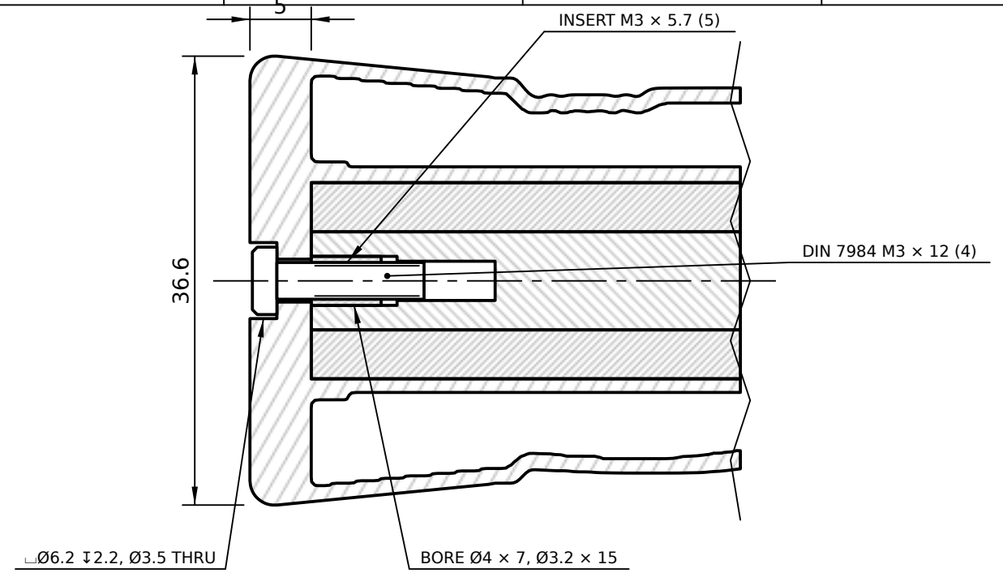
SECTION G–G

SCALE 2:1 · HEAD, IN THE SLIT ZONE (y = 120)



FIT-TEST COUPON (fit_test_slot.stl)

SCALE 1:1 · 8 THK · SLOTS 15.6 / 16.0 / 16.4 × 30.2



SECTION F–F (BUTT END)

SCALE 2:1

ITEM	PART	MATERIAL / SPECIFICATION	QTY	MASS g EL / HY
1	Face sheet	Woven carbon/epoxy plate 4 thk, cut to the profile DXF	2	304 / 284
2	Rib support	PA-CF printed, 1 or 2 pieces – sheet 3 of PB-EL-001 / PB-HY-001	1	178 / 170
3	Grip	PLA printed, any rev-8 design (drawn CARBON MATRIX), 60–65 g	1	61
4	Screw	DIN 7984 M3 × 12 low-head cap screw, A2 stainless	1	1.0
5	Threaded insert	M3 heat-set, brass, 5.7 long, in a Ø4 bore	1	0.4
6	Adhesive	Structural epoxy (rib support to face sheets)	a/r	–
COMPLETE PADDLE WITH GRIP (3D MODEL)				**849 / 802**

NOTES

1. GRIP (3): ANY OF THE TEN REV-8 PRINTED GRIPS (pickleball_grips) - SAME INTERFACE, DIFFERENT SURFACE; DRAWN: CARBON MATRIX. PLA 60–65 g (61.2 g DRAWN). PRINT UPRIGHT, BUTT DOWN, 100 % INFILL, NO SUPPORTS.
2. SLOT 30.2 × 16 × 130 DEEP, 1 LEAD-IN; HANDLE 30 × 16 NOMINAL. THE BONDED 4/8/4 STACK MAY MEASURE 16.1–16.5: MEASURE IT, PRINT THE FIT-TEST COUPON AND SET SLOT_T (pickleball_grips/scripts/grip_core.py) TO SUIT.
3. END WALLS SLIT OVER THE TOP 20 (y 115–135) SO THE THROAT FLARE (FROM y = 130) PASSES; THE GRIP SEATS ON THE BODY BUTT AT y = 5 (= GRIP BASE 5).
4. INSERT (5): HEAT-SET M3 × 5.7 INTO THE Ø4 BORE, FLUSH WITH THE CORE BUTT, BEFORE THE FACE SHEETS ARE BONDED.
5. SCREW (4): DIN 7984 M3 × 12 LOW-HEAD (M3 × 16 ALSO FITS), LOW-STRENGTH THREADLOCKER, HAND-TIGHT (≈0.5 N·m) SO THE PLA BASE DOES NOT CRACK; 2.8 OF FLOOR UNDER THE HEAD.
6. OVERALL LENGTH 390 INCL. THE 5 GRIP BASE; HANDLE 130 FROM THE BUTT TO THE THROAT. MASS WITH GRIP 849 / 802 g (EL / HY).
7. SAME GRIP ON PB-EL-001 AND PB-HY-001. ALL DIMENSIONS IN mm.

PRINTED GRIP – PADDLE INTERFACE

PRINTED GRIP 135 LONG, SLOT 30.2 × 16 · M3 LOW-HEAD SCREW + HEAT-SET INSERT · BOTH PADDLES

DRAWN	DATE	CHECKED	APPROVED
Claude Code	2026-10-24		

DRAWING No.	REV	SHEET	SIZE	SCALE	UNITS
PB-GR-002	C	1 / 1	A3	1:1, 2:1	mm

PROJECTION	MATERIAL	STATUS
	SEE BOM	ISSUED FOR REVIEW